$$\vec{B} = \int d\vec{B} = \frac{\mu_\circ I}{4\pi} \int \frac{d\vec{l} \times \hat{r}}{r^2} \qquad (3.35)$$

where the integral is taken over the entire current distribution.

Cases

1. If the point P lies on the conductor, then $\theta = 0^\circ$. Therefore, $|d\vec{B}|$ is zero.
2. If the point lies perpendicular to the conductor, then $\theta = 90^\circ$. Therefore, $d\vec{B}$ is maximum and is given by $d\vec{B} = \frac{\mu_\circ}{4\pi} \frac{I\,dl}{r^2} \hat{n}$ where $\hat{n}$ is the unit vector perpendicular to both $I\,d\vec{l}$ and $\hat{r}$

Note

Electric current is not a vector quantity. It is a scalar quantity. But electric current in a conductor has direction of flow. Therefore, the electric current flowing in a small elemental conductor can be taken as vector quantity i.e. $Id\vec{l}$

Similarities between electric field (from Coulomb's law) and magnetic field (from Biot-Savart's law)

Electric and magnetic fields

- obey inverse square law, so they are long range fields.
- obey the principle of superposition and are linear with respect to source. In magnitude,

$$E \propto q$$
$$B \propto Idl$$

Differences between electric field (from Coulomb's law) and magnetic field (from Biot-Savart's law)

S. No.	Electric field	Magnetic field
1	Produced by a scalar source i.e., an electric charge q	Produced by a vector source i.e., current element $I\,d\vec{l}$
2	It is directed along the position vector joining the source and the point at which the field is calculated	It is directed perpendicular to the position vector $\vec{r}$ and the current element $Id\vec{l}$
3	Does not depend on angle	Depends on the angle between the position vector $\vec{r}$ and the current element $Id\vec{l}$

Note

The exponent of charge q (source) and exponent of electric field E is unity. Similarly, the exponent of current element Idl (source) and exponent of magnetic field B is unity. In other words, electric field $\vec{E}$ is proportional only to charge (source) and not on higher powers of charge (q^2, q^3, etc). Similarly, magnetic field $\vec{B}$ is proportional to current element $Id\vec{l}$ (source) and not on square or cube or higher powers of current element. The cause and effect have linear relationship.

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